Advanced Cisco Networking Academy: Designing a Multiprotocol Network with Internal/External BGP

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Purpose
The purpose of this lab is to expand our knowledge on the BGP routing protocol by teaching the internal variation of the protocol. It adds an extra layer of difficulty by having 3 different remote autonomous systems that run their own routing protocol and need to be connected by BGP. This lab also tests previously thought skills as how to implement eBGP and how to connect BGP to other routing protocols such as OSPF, IS-IS, and EIGRP.

Background
Border Gateway Protocol, or BGP, is a routing protocol used to exchange routers in between routers on internal networks and on the internet. It was developed in 1989 by the Internet Engineering Task Force, or IETF, a group that develops standards and operates under the Internet Society non-profit organization. BGP for IPv4 was originally specified in RFC 1105, and BGP for IPv6 was originally specified in RFC 1654. The most recent version of the protocol, BGP-4, is specified in RFC 4271.

BGP uses TCP connections between routers on port 179 and sends hello messages every 30 seconds by default. Internal BGP, or iBGP, sends routing information between routers in the same autonomous system, while external BGP, or eBGP, sends information between autonomous systems. Confusingly, eBGP sounds very similar to EGP, or Exterior Gateway Protocol, which is the predecessor to BGP. BGP is mainly designed to allow routers managed by different companies to exchange routes with each other, which is why eBGP is much more commonly used than iBGP.

A helpful characteristic of BGP is that BGP-enabled routers can peer with each other even if they aren't directly connected. If the routers have a route to each other, they can become BGP neighbors irrelevant of the number of hops between them.

BGP routers go through 6 exchange states: Idle, Connect, Active, OpenSent, OpenConfirm, and Established. By default, if the protocol is given multiple routes to a destination, it will count the number of AS jumps to the destination between each route and use the route with the lowest number.

One of BGP's major vulnerabilities is that routers, by default, BGP-enabled routers will put any advertised routes into the routing table. While this is normally useful, it also leaves routers vulnerable to hijacking, and makes it somewhat trivial for a malicious actor to redirect traffic to their own servers by introducing a lower-metric route

Open Shortest Path First is a routing protocol that uses link state advertisements to automatically build a network topology and provide end-to-end connectivity across networks. It was developed by the IETF. This lab uses both OSPFv2 and OSPFv3, which operate similarly but uses IPv4 and IPv6 addresses respectively. OSPFv2 is specified in RFC 2328 and OSPFv3 is specified in RFC 5320.

OSPF-enabled routers can have two different types of relationships with each other: neighboring relationships and adjacencies. Neighbor relationships, which are automatically formed in P2P, broadcast, and point-to-multipoint networks, are formed using hello messages and simply inform routers of each other's existence on the network. Adjacencies are a more complex type of relationship that allows routers to exchange routing information with each other.

Intermediate System to Intermediate System, or IS-IS, is an interior gateway linkstate routing protocol. Unlike most routing protocols, it was standardized by the International Standards Organization instead of the IETF. More specifically, it was defined in ISO/IEC 10589:2002. IS-IS can send 4 types of packets: Hello PDUs, Link State PDUs, Complete Sequence Number PDUs, and Partial Sequence Number PDUs. IS-IS uses Dijkstra's algorithm, like OSPF, to find the optimal routes through a network.

Enhanced Interior Gateway Routing Protocol, or EIGRP, is a proprietary routing protocol developed by Cisco. In 2013, the EIGRP protocol was made partially open, as Cisco released documentation for a stripped-down version of the protocol for use with other vendors' routers. Unlike OSPF and IS-IS, EIGRP uses the diffusing update algorithm, or DUAL, to determine the best path between routers. EIGRP uses the term "successor" to describe the best next-hop route to a destination and keeps a log of feasible successors to use in the case that a successor goes down.

Lab Summary

In this lab, we configured 3 remote autonomous systems, one running OSPF, one running EIGRP, and the other runner ISIS to be connected via BGP. For this lab, we had to redistribute routes learned from one routing protocol into the BGP domain. The final goal was to set up connectivity from one endpoint to the other/ The network had full mesh connectivity, and we tested it by the tracert command between two routers on the opposite end of the network. We also added complexity by adding two tuning features to the BGP network, like weight and local preference.

Lab Commands

Configuring BGP (Basic)

```
Router(config)#router bgp <as>
```

Enters BGP configuration mode. <as> represents the router's BGP autonomous system number.

```
Router(config-router)#bgp router-id <id>
```

Configures the router's BGP router ID.

```
Router(config-router)#neighbor <ip> remote-as <remote-as>
```

Configures a BGP neighbor. IP can be IPv4 or IPv6. <remote-as> is the autonomous system number of the neighbor. If the <remote-as> number matches the local AS number, iBGP is being configured. Otherwise, eBGP is being configured.

```
Router(config-router)#neighbor <ip> description <desc>
```

Configures a description for a neighbor.

```
Router(config-router)#address-family [ipv4|ipv6]
```
Enters address family configuration mode for the specified IP version.

```
Router(config-router-af)#network <network> mask <subnet-mask>
```
Configures the network where a neighbor can be found.

```
Router(config-router-af)#redistribute connected subnets
```
Redistributes networks from directly connected interfaces into BGP.

```
Router(config-router-af)#redistribute ospf <pid> metric <metric>
```
Redistributes information into BGP from OSPF using the given process ID. Redistributed routes will be given the specified metric.

```
Router(config-router-af)#redistribute isis <tag> [level-1|level-2|level-1-2] metric <metric>
```
Redistributes information into BGP from IS-IS using the given area tag. Routes from level 1 (intra-area), level 2 (inter-area), or both levels can be redistributed. Redistributed routes will be given the specified metric.

```
Router(config-router-af)#neighbor <ip> activate
```
Activates a neighbor in address family configuration mode.

```
Router(config-router-af)#distance bgp <ead> <iad> <lad>
```
Changes the administrative distance for routes learned via BGP. <ead>, <iad>, and <lad> correspond to the distance for external, internal, and local BGP routes respectively.

Configuring EIGRP

```
Router(config)#ipv6 router eigrp <as>
```
Enters EIGRP router configuration mode using the specified autonomous system number. IPv4 and IPv6 have separate router configuration modes.

```
Router(config-router)#eigrp router-id <x.x.x.x>
```
Sets an EIGRP router ID.

```
Router(config-router)#network <x.x.x.x>
```
Specifies a network to be advertised into EIGRP. This command only works for IPv4 addresses.

```
Router(config-if)#ipv6 eigrp <as>
```
Specifies an interface's network to be advertised into EIGRP. This command only works for IPv6 addresses.

Configuring OSPF

```
Router(config)#ipv6 router ospf <pid>
```
Enters OSPF router configuration mode using the specified process ID. IPv4 and IPv6 have separate router configuration modes.

```
Router(config-router)#router-id <id>
```
Configures an OSPF router ID.

```
Router(config-router)#redistribute bgp <as> subnets
```
Redistributes routes into OSPF from BGP using the specified autonomous system number.

```
Router(config-if)#ip ospf <pid> area <area>
```
Configures OSPF on an interface using the specified process ID and area.

```
Router(config-if)#ipv6 ospf <pid> area <area>
```
Configures OSPF on an interface using the specified process ID and area.

Configuring IS-IS

```
Router(config)#router isis <tag>
```
Enters IS-IS configuration mode with the specified area tag.

```
Router(config-router)#net <net>
```
Configures an IS-IS Network Entity Title. This parameter works similarly to a router ID in other routing protocols.

```
Router(config-router)#metric-style wide
```
Configures IS-IS to use larger metrics in best-path calculations.

```
Router(config-router)#redistribute bgp <as>
```
Redistributes into IS-IS from the given BGP autonomous system number.

Configuring BGP (Route Map)

```
Router(config)#ip prefix-list <list> seq <number> [permit|deny]
<ip>/<mask>
```
Configures a prefix list with the name <list> to permit or deny a certain range of IP addresses. Since a prefix list can consist of multiple lines, the sequence number determines the order in which the lines are read.

```
Router(config)#route-map <map-name> [permit|deny] <seq-number>
```
Configures a route map with the name <map-name> to permit or deny addresses. The sequence number determines the order in which the route map instructions are read.

```
Router(config-route-map)#match ip address prefix-list <prefix-
list>
```

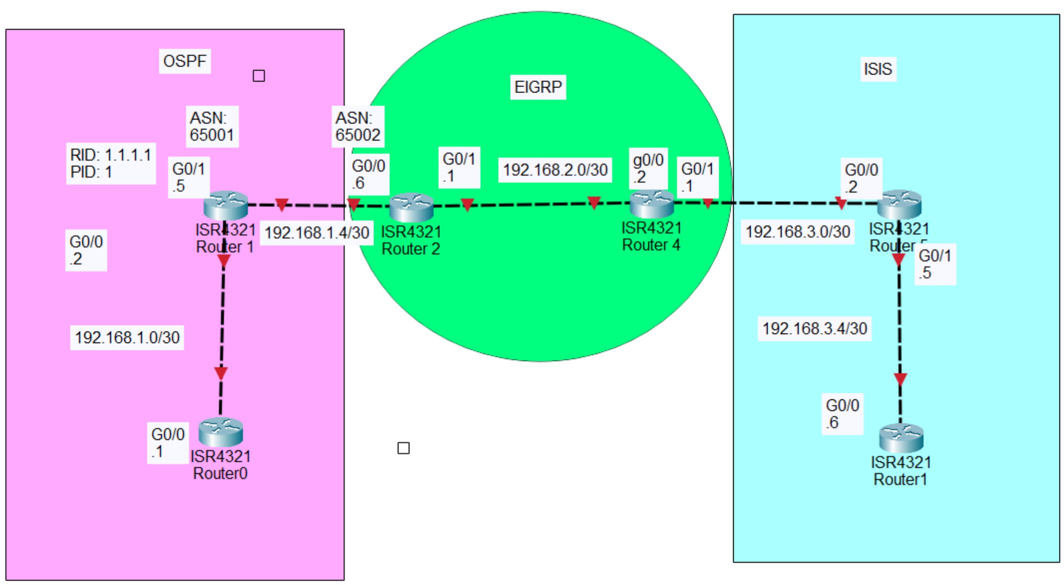
Sets a route map to permit/deny addresses specified in a prefix list.

```
Router(config-router-af)#neighbor <ip> route-map <route-map>
[in|out]
```

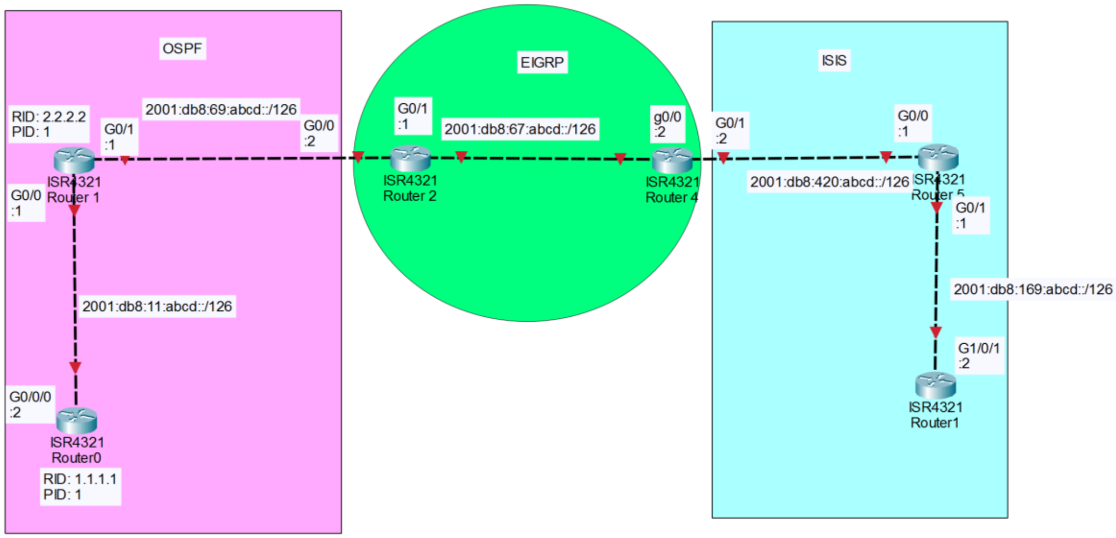
Sets the routes given to/from a BGP neighbor to be passed through a specific route map for filtering/manipulation.

Network Diagrams

iPv4 Network Topology



IPV6 Network Topology



Lab Configurations

R1:

```
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname R1
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
login on-success lo
subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing multilink bundle-name
spanning-tree extend system-id
Redundancy
 mode non
interface GigabitEthernet0/0/0
 ip address 192.168.1.1 255.255.255.252
 ip ospf 1 area 0
```

```
 negotiation auto
 ipv6 address 2001:DB8:11:ABCD::2/126
 ipv6 ospf 1 area 0
interface GigabitEthernet0/0/1
 no ip address
 negotiation auto
interface GigabitEthernet0/2/0
 no ip address
 negotiation auto
interface GigabitEthernet0/2/1
 no ip address
 negotiation auto
interface GigabitEthernet0
vrf forwarding Mgmt-intf
 no ip address
 negotiation auto
router ospf 1
 router-id 1.1.1.1
 ip forward-protocol nd
 no ip http server
 ip http secure-server
ipv6 router ospf 1
 router-id 1.1.1.1
control-plane
line con 0
 transport input none
 stopbits 1
line aux 0
 stopbits 1
line vty 0 4
login
end
```

R2:

```
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname R2
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
 address-family ipv4
 exit-address-family
```

```
 address-family ipv6
 exit-address-family
no aaa new-model
login on-success log
subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FLM24060912
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
 mode none
interface GigabitEthernet0/0/0
 ip address 192.168.1.2 255.255.255.252
 ip ospf 1 area 0
 negotiation auto
 ipv6 address 2001:DB8:11:ABCD::1/126
 Ipv6 ospf 1 area 0
interface GigabitEthernet0/0/1
 ip address 192.168.1.5 255.255.255.252
 negotiation auto
 ipv6 address 2001:DB8:69:ABCD::1/126
interface GigabitEthernet0/2/0
 no ip address
 negotiation auto
interface GigabitEthernet0/2/1
 no ip address
 negotiation auto
interface GigabitEthernet0
 vrf forwarding Mgmt-intf
 no ip address
 negotiation auto
router ospf 1
 router-id 2.2.2.2
 redistribute connected subnets
 redistribute bgp 65001 subnets
router bgp 65001
 bgp log-neighbor-changes
 neighbor 2001:DB8:69:ABCD::2 remote-as 65002
 neighbor 192.168.1.6 remote-as 65002
 address-family ipv4
  redistribute connected
  redistribute ospf 1
  no neighbor 2001:DB8:69:ABCD::2 activate
```

```
  neighbor 192.168.1.6 activate
exit-address-family
address-family ipv6
  redistribute connected
  redistribute ospf 1
  neighbor 2001:DB8:69:ABCD::2 activate
exit-address-family
ip forward-protocol nd
no ip http server
ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router ospf 1
 router-id 2.2.2.2
 redistribute connected
 redistribute bgp 65001
control-plane
line con 0
 transport input none
 stopbits 1
line aux 0
 stopbits 1
line vty 0 4
login
end
```

R3:

```
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname r3
boot-start-marker
boot system bootflash:isr4300-universalk9.16.09.08.SPA.bin
boot-end-marker
vrf definition Mgmt-intf
 address-family ipv4
 exit-address-family
 Address-family ipv6
 exit-address-family
no aaa new-model
login on-success log
subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
multilink bundle-name authenticated
```

```
license udi pid ISR4321/K9 sn FDO214420G7
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
interface Loopback0
 description eigrp test
 ip address 3.3.3.3 255.255.255.255
interface GigabitEthernet0/0/0
 description bgp
 ip address 192.168.1.6 255.255.255.252
 negotiation auto
 ipv6 address 2001:DB8:69:ABCD::2/126
interface GigabitEthernet0/0/1
 description eigrp
 Bandwidth 100000
 ip address 192.168.2.1 255.255.255.252
 negotiation auto
 ipv6 address 2001:DB8:67:ABCD::1/126
 ipv6 eigrp 100
interface Serial0/1/0
 no ip address
 shutdown
interface Serial0/1/1
 no ip address
 shutdown
interface GigabitEthernet0
 vrf forwarding Mgmt-intf
 no ip address
 shutdown
 negotiation auto
router eigrp 100
 network 3.3.3.3 0.0.0.0
 network 4.4.4.4 0.0.0.0
 network 192.168.2.0 0.0.0.3
 redistribute bgp 65002 metric 10000 100 255 1    1500
 redistribute connected
 passive-interface default
 no passive-interface GigabitEthernet0/0/1
 no passive-interface Loopback0
router bgp 65002
 bgp log-neighbor-changes
 neighbor 2001:DB8:69:ABCD::1 remote-as 65001
 neighbor 192.168.1.5 remote-as 65001
 neighbor 192.168.2.2 remote-as 65002
address-family ipv4
```

```
  redistribute connected
  redistribute eigrp 100
  no neighbor 2001:DB8:69:ABCD::1 activate
  neighbor 192.168.1.5 activate
  neighbor 192.168.1.5 weight 500
  neighbor 192.168.2.2 activate
  neighbor 192.168.2.2 weight 300
exit-address-family
address-family ipv6
  redistribute connected
  redistribute eigrp 100
  neighbor 2001:DB8:69:ABCD::1 activate
exit-address-family
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
ip route 192.168.2.2 255.255.255.255 192.168.2.2 90
ipv6 router eigrp 100
 redistribute connected
 redistribute bgp 65002 metric 10000 100 255 1 1500
route-map BGP-TO-EIGRP permit 10
set metric 1000000 10 255 1 1500
set tag 52
route-map EIGRP-TO-BGP deny 5
match tag 52
control-plane
line con 0
 transport input none
 stopbits 1
line aux 0
 stopbits 1
line vty 0 4
login
end
```

R4

```
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname r4
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
```

```
 address-family ipv4
 exit-address-family
 address-family ipv6
 exit-address-family
no aaa new-model
login on-success log
subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FDO211216BL
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
interface Loopback0
 description eigrp test
 ip address 4.4.4.4 255.255.255.255
interface GigabitEthernet0/0/0
 description eigrp
 bandwidth 100000
 ip address 192.168.2.2 255.255.255.252
 negotiation auto
 ipv6 address 2001:DB8:67:ABCD::2/126
 ipv6 eigrp 100
interface GigabitEthernet0/0/1
 description bgp
 ip address 192.168.3.1 255.255.255.252
 negotiation auto
 ipv6 address 2001:DB8:420:ABCD::2/126
 interface Serial0/1/0
 no ip address
interface Serial0/1/1
 no ip address
interface GigabitEthernet0
 vrf forwarding Mgmt-intf
 no ip address
 negotiation auto
router eigrp 100
 network 3.3.3.3 0.0.0.0
 network 4.4.4.4 0.0.0.0
 network 192.168.2.0 0.0.0.3
 redistribute bgp 65002 metric 10000 100 255 1  1500
no passive-interface GigabitEthernet0/0/0
no passive-interface Loopback0
```

```
router bgp 65002
 bgp log-neighbor-changes
 neighbor 2001:DB8:420:ABCD::1 remote-as 65003
 neighbor 192.168.2.1 remote-as 65002
 neighbor 192.168.3.2 remote-as 65003
address-family ipv4
  redistribute eigrp 100
  no neighbor 2001:DB8:420:ABCD::1 activate
  neighbor 192.168.2.1 activate
  neighbor 192.168.2.1 weight 300
  neighbor 192.168.3.2 activate
  neighbor 192.168.3.2 weight 500
  neighbor 192.168.3.2 route-map PREF=ISIS in
exit-address-family
address-family ipv6
  redistribute eigrp 100
  neighbor 2001:DB8:420:ABCD::1 activate
exit-address-family
ip forward-protocol nd
no ip http server
ip http secure-server
ipv6 router eigrp 100
redistribute connected
redistribute bgp 65002 metric 10000 100 255 1 1500
route-map PREF=ISIS permit 10 set local-preference 300
control-plane
line con 0
 transport input none
 stopbits 1
line aux 0
 stopbits 1
line vty 0 4
 login
end
```

R5

```
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname R5
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
 address-family ipv4
 exit-address-family
```

```
 address-family ipv6
 exit-address-family
no aaa new-model
login on-success log
subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FDO214421CF
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
interface Loopback0
 ip address 5.5.5.5 255.255.255.255
 ip router isis 1
interface GigabitEthernet0/0/0
 ip address 192.168.3.2 255.255.255.252
 negotiation auto
 ipv6 address 2001:DB8:420:ABCD::1/126
interface GigabitEthernet0/0/1
 ip address 192.168.3.5 255.255.255.252
 ip router isis 1
 negotiation auto
 ipv6 address 2001:DB8:169:ABCD::1/126
interface Serial0/1/0
 no ip address
interface Serial0/1/1
 no ip address
interface GigabitEthernet0
 vrf forwarding Mgmt-intf
 no ip address
 negotiation auto
router isis 1
 net 49.0001.0000.0000.0005.00
 is-type level-1
 metric-style wide
 redistribute connected
 redistribute bgp 65003 metric 1 level-1
address-family ipv6
  redistribute connected
  redistribute bgp 65003
exit-address-family
router bgp 65003
 bgp log-neighbor-changes
```

```
 neighbor 2001:DB8:420:ABCD::2 remote-as 65002
 neighbor 192.168.3.1 remote-as 65002
 address-family ipv4
  redistribute connected
  redistribute isis 1 level-1
  no neighbor 2001:DB8:420:ABCD::2 activate
  neighbor 192.168.3.1 activate
exit-address-family
address-family ipv6
  redistribute connected
  redistribute isis 1 level-1
  network 2001:DB8:67:ABCD::/126
  neighbor 2001:DB8:420:ABCD::2 activate
exit-address-family
ip forward-protocol nd
control-plane
line con 0
 transport input none
 stopbits 1
line aux 0
 stopbits 1
line vty 0 4
Login
End
```

R6

```
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname R6
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
login on-success log
subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
multilink bundle-name authenticated
```

```
license udi pid ISR4321/K9 sn FDO21441WBL
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
interface Loopback0
 ip address 6.6.6.6 255.255.255.255
 ip router isis 1
interface GigabitEthernet0/0/0
 ip address 192.168.3.6 255.255.255.252
 ip router isis 1
 shutdown
 negotiation auto
 ipv6 address 2001:DB8:168:ABCD::2/126
interface GigabitEthernet0/0/1
 no ip address
 shutdown
 negotiation auto
interface Serial0/1/0
 no ip address
 shutdown
interface Serial0/1/1
 no ip address
 shutdown
interface GigabitEthernet0/2/0
 no ip address
 shutdown
 negotiation auto
interface GigabitEthernet0/2/1
 no ip address
 shutdown
 negotiation auto
interface GigabitEthernet0
 vrf forwarding Mgmt-intf
 no ip address
 shutdown
 negotiation auto
router isis 1
 net 49.0001.0000.0000.0006.00
 is-type level-1
 metric-style wide
router isis
 metric-style narrow

ip forward-protocol nd
ip http server
```

```
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
control-plane
line con 0
 transport input none
 stopbits 1
line aux 0
 stopbits 1
line vty 0 4
 login
End
```

Testing :

TraceRoute through the network showing full connectivity mesh throughout the network

```
R1#traceroute 192.168.3.6
Tracing the route to 192.168.3.6
VRF info: (vrf in name/id, vrf out name/id)
  1 192.168.1.2 1 msec 1 msec 0 msec
  2 192.168.1.6 1 msec 1 msec 1 msec
  3 192.168.2.2 1 msec 1 msec 1 msec
  4 192.168.3.2 1 msec 2 msec 1 msec
  5 192.168.3.6 1 msec 2 msec *
```

```
R6#traceroute 192.168.1.1
Tracing the route to 192.168.1.1
VRF info: (vrf in name/id, vrf out name/id)
  1 192.168.3.5 2 msec 1 msec 0 msec
  2 192.168.3.1 0 msec 1 msec 1 msec
  3 192.168.2.1 1 msec 1 msec 1 msec
  4 192.168.1.5 2 msec 1 msec 1 msec
  5 192.168.1.1 2 msec 2 msec *
```

Routing tables for each router showing the BGP and redistribution routes.

R1:

```
      3.0.0.0/32 is subnetted, 1 subnets
O E2     3.3.3.3 [110/1] via 192.168.1.2, 23:10:38,
GigabitEthernet0/0/0
      4.0.0.0/32 is subnetted, 1 subnets
```

```
O E2     4.4.4.4 [110/1] via 192.168.1.2, 23:10:09,
GigabitEthernet0/0/0
      5.0.0.0/32 is subnetted, 1 subnets
O E2     5.5.5.5 [110/1] via 192.168.1.2, 23:04:38,
GigabitEthernet0/0/0
      6.0.0.0/32 is subnetted, 1 subnets
O E2     6.6.6.6 [110/1] via 192.168.1.2, 00:07:47,
GigabitEthernet0/0/0
      192.168.1.0/24 is variably subnetted, 3 subnets, 2 masks
C        192.168.1.0/30 is directly connected,
GigabitEthernet0/0/0
L        192.168.1.1/32 is directly connected,
GigabitEthernet0/0/0
O E2     192.168.1.4/30
           [110/20] via 192.168.1.2, 23:10:50,
GigabitEthernet0/0/0
      192.168.2.0/30 is subnetted, 1 subnets
O E2     192.168.2.0 [110/1] via 192.168.1.2, 23:10:38,
GigabitEthernet0/0/0
      192.168.3.0/30 is subnetted, 1 subnets
O E2     192.168.3.4 [110/1] via 192.168.1.2, 00:08:17,
GigabitEthernet0/0/0
```

R2:

```
      3.0.0.0/32 is subnetted, 1 subnets
B        3.3.3.3 [20/0] via 192.168.1.6, 23:17:57
      4.0.0.0/32 is subnetted, 1 subnets
B        4.4.4.4 [20/153856] via 192.168.1.6, 23:17:27
      5.0.0.0/32 is subnetted, 1 subnets
B        5.5.5.5 [20/281856] via 192.168.1.6, 23:11:56
      6.0.0.0/32 is subnetted, 1 subnets
B        6.6.6.6 [20/281856] via 192.168.1.6, 00:15:05
      192.168.1.0/24 is variably subnetted, 4 subnets, 2 masks
C        192.168.1.0/30 is directly connected,
GigabitEthernet0/0/0
L        192.168.1.2/32 is directly connected,
GigabitEthernet0/0/0
C        192.168.1.4/30 is directly connected,
GigabitEthernet0/0/1
L        192.168.1.5/32 is directly connected,
GigabitEthernet0/0/1
      192.168.2.0/30 is subnetted, 1 subnets
B        192.168.2.0 [20/0] via 192.168.1.6, 23:17:57
      192.168.3.0/30 is subnetted, 1 subnets
B        192.168.3.4 [20/281856] via 192.168.1.6, 00:15:35
```

R3:

```
      3.0.0.0/32 is subnetted, 1 subnets
C        3.3.3.3 is directly connected, Loopback0
      4.0.0.0/32 is subnetted, 1 subnets
D        4.4.4.4 [90/153856] via 192.168.2.2, 23:19:28,
GigabitEthernet0/0/1
      5.0.0.0/32 is subnetted, 1 subnets
D EX     5.5.5.5 [170/281856] via 192.168.2.2, 23:13:39,
GigabitEthernet0/0/1
      6.0.0.0/32 is subnetted, 1 subnets
D EX     6.6.6.6 [170/281856] via 192.168.2.2, 00:16:47,
GigabitEthernet0/0/1
      192.168.1.0/24 is variably subnetted, 3 subnets, 2 masks
B        192.168.1.0/30 [20/0] via 192.168.1.5, 23:19:40
C        192.168.1.4/30 is directly connected,
GigabitEthernet0/0/0
L        192.168.1.6/32 is directly connected,
GigabitEthernet0/0/0
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.2.0/30 is directly connected,
GigabitEthernet0/0/1
L        192.168.2.1/32 is directly connected,
GigabitEthernet0/0/1
      192.168.3.0/30 is subnetted, 1 subnets
D EX     192.168.3.4
           [170/281856] via 192.168.2.2, 00:17:17,
GigabitEthernet0/0/1
```

R4:

```
      3.0.0.0/32 is subnetted, 1 subnets
D        3.3.3.3 [90/153856] via 192.168.2.1, 23:19:55,
GigabitEthernet0/0/0
      4.0.0.0/32 is subnetted, 1 subnets
C        4.4.4.4 is directly connected, Loopback0
      5.0.0.0/32 is subnetted, 1 subnets
B        5.5.5.5 [20/0] via 192.168.3.2, 23:14:10
      6.0.0.0/32 is subnetted, 1 subnets
B        6.6.6.6 [20/20] via 192.168.3.2, 00:17:17
      192.168.1.0/30 is subnetted, 2 subnets
D EX     192.168.1.0
           [170/281856] via 192.168.2.1, 23:19:55,
GigabitEthernet0/0/0
D EX     192.168.1.4
```

```
           [170/26112] via 192.168.2.1, 23:19:55, 
GigabitEthernet0/0/0
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.2.0/30 is directly connected, 
GigabitEthernet0/0/0
L        192.168.2.2/32 is directly connected, 
GigabitEthernet0/0/0
      192.168.3.0/24 is variably subnetted, 3 subnets, 2 masks
C        192.168.3.0/30 is directly connected, 
GigabitEthernet0/0/1
L        192.168.3.1/32 is directly connected, 
GigabitEthernet0/0/1
B        192.168.3.4/30 [20/0] via 192.168.3.2, 00:17:47
```

R5:

```
      3.0.0.0/32 is subnetted, 1 subnets
B        3.3.3.3 [20/153856] via 192.168.3.1, 23:17:27
      4.0.0.0/32 is subnetted, 1 subnets
B        4.4.4.4 [20/0] via 192.168.3.1, 23:17:27
      5.0.0.0/32 is subnetted, 1 subnets
C        5.5.5.5 is directly connected, Loopback0
      6.0.0.0/32 is subnetted, 1 subnets
i L1     6.6.6.6 [115/20] via 192.168.3.6, 00:21:05, 
GigabitEthernet0/0/1
      192.168.1.0/30 is subnetted, 2 subnets
B        192.168.1.0 [20/281856] via 192.168.3.1, 23:17:27
B        192.168.1.4 [20/26112] via 192.168.3.1, 23:17:27
      192.168.2.0/30 is subnetted, 1 subnets
B        192.168.2.0 [20/0] via 192.168.3.1, 23:17:27
      192.168.3.0/24 is variably subnetted, 4 subnets, 2 masks
C        192.168.3.0/30 is directly connected, 
GigabitEthernet0/0/0
L        192.168.3.2/32 is directly connected, 
GigabitEthernet0/0/0
C        192.168.3.4/30 is directly connected, 
GigabitEthernet0/0/1
L        192.168.3.5/32 is directly connected, 
GigabitEthernet0/0/1
```

R6:

```
      3.0.0.0/32 is subnetted, 1 subnets
i L1     3.3.3.3 [115/11] via 192.168.3.5, 00:22:00, 
GigabitEthernet0/0/0
      4.0.0.0/32 is subnetted, 1 subnets
```

```
i L1     4.4.4.4 [115/11] via 192.168.3.5, 00:22:00,
GigabitEthernet0/0/0
      5.0.0.0/32 is subnetted, 1 subnets
i L1     5.5.5.5 [115/20] via 192.168.3.5, 00:22:00,
GigabitEthernet0/0/0
      6.0.0.0/32 is subnetted, 1 subnets
C        6.6.6.6 is directly connected, Loopback0
      192.168.1.0/30 is subnetted, 2 subnets
i L1     192.168.1.0 [115/11] via 192.168.3.5, 00:22:00,
GigabitEthernet0/0/0
i L1     192.168.1.4 [115/11] via 192.168.3.5, 00:22:00,
GigabitEthernet0/0/0
      192.168.2.0/30 is subnetted, 1 subnets
i L1     192.168.2.0 [115/11] via 192.168.3.5, 00:22:00,
GigabitEthernet0/0/0
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.3.4/30 is directly connected,
GigabitEthernet0/0/0
L        192.168.3.6/32 is directly connected,
GigabitEthernet0/0/0
```

Other Show Commands:

Shows the Weight and Local Preference tuning for BGP:

```
r4#show bgp topology *

BGP table version is 12, local router ID is 4.4.4.4

     Network          Next Hop            Metric LocPrf Weight
Path
* i  3.3.3.3/32       192.168.2.1              0    100    300 ?
*>                    192.168.2.1         153856         32768 ?
*>   4.4.4.4/32       0.0.0.0                  0         32768 ?
*>   5.5.5.5/32       192.168.3.2              0    300    500
65003 ?
*>   6.6.6.6/32       192.168.3.2             20    300    500
65003 ?
* i  192.168.1.0/30   192.168.1.5              0    100    300
65001 ?
*>                    192.168.2.1         281856         32768 ?
* i  192.168.1.4/30   192.168.2.1              0    100    300 ?
*>                    192.168.2.1          26112         32768 ?
* i  192.168.2.0/30   192.168.2.1              0    100    300 ?
*>                    0.0.0.0                  0         32768 ?
```

```
     Network          Next Hop            Metric LocPrf Weight
Path
r>   192.168.3.0/30   192.168.3.2              0    300    500
65003 ?
*>   192.168.3.4/30   192.168.3.2              0    300    500
65003 ?
```

Problems

Interface Statements on EIGRP

- We tried to configure EIGRP in IPv4 with interface statements. As this behavior is only permitted for IPv6 EIGRP, we had to replace the `ip eigrp <as>` command with the `network <x.x.x.x>` command.

EIGRP Information requirement

- We originally tried to redistribute from BGP into EIGRP with the command redistribute `bgp <as> metric <metric>.` This raised an error, as EIGRP requires much more information than this to redistribute routes. The actual command looks like this: `redistribute bgp <as> metric <metric> <delay> <reliability> <bandwidth> <mtu>`, like this (`Redistribute bgp 2 metric 100 100 255 255 1)`

Redistribution

- We couldn't ping throughout the network because the redistribution was only one way. The OSPF Routes were redistributed into the BGP and then into the EIGRP and into the ISIS, but backwards, there were holes on where the redistribution was occurring. We figured this out by looking at the routing tables for each router and seeing that there were a lot of ISIS routes and EIGRP routes missing on routers 3 and 4. We typed in the redistribution commands, and it worked.

IPV6 connectivity

- On each BGP and EIGRP, we had to have a separate Address family for IPV6, and that was not configured on some of the routers initially, so our IPV6 trace routes were not working. We looked at the routing tables and figured out that some of the routers in the EIGRP network didn't redistribute in IPV6 and only on IPV4.

ISIS

- When working with IS-IS we used the same commands, but the pings failed. It took some research to learn that the redistribution commands default to IS-IS level 2, whereas our IS-IS network used level 1. After manually assigning level-2, everything worked.

Conclusion

This lab was an excellent introduction to external BGP and how to connect different networks running with different protocols. It taught us multiple technical skills including redistribution of different routes and also configuring new protocols like IS-IS and Eigrp. ’m

hopeful that this more complete understanding of both external and internal BGP, and how they communicate with both Cisco-proprietary and open
Standard routing protocols will aid me in my networking ability in the future. By far, the most challenging part of this lab was keeping iBGP routes working between R3
and R4 while eliminating routing loops, but figuring out how to fix this problem
using a route map was by far the most satisfying part of the lab, greatly improving my
Understanding the optional characteristics of BGP.